Caller ⇄ SmartPRS2 ⇄ SmartEPT — Field-Force Integration Report

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Summary

Caller (Ametecs Managed Mobile), SmartPRS2 and SmartEPT are now connected through API-key-authenticated REST APIs, built on 6 Oct 2026, and all three test suites pass.

- **Geofenced sign-in:** a field employee can sign in to the Caller app only inside the radius set in SmartPRS2 (Field Force → Geofence Rule).
- **Attendance from app login/logout:** each app login and logout becomes an IN/OUT punch, with location, in both SmartPRS2 and SmartEPT.
- **Field trail and enforcement:** every GPS fix goes to SmartEPT. Leaving the fence raises a `GEOFENCE_EXIT` violation in SmartEPT's existing violations feed, and returning resolves it.
- **Additive only:** no existing screen, table or flow in SmartPRS2 or SmartEPT changed behaviour. SmartPRS2 got 3 new files and 2 route lines. SmartEPT got 2 new files, 4 optional fields on an existing endpoint and 2 route lines.

Who owns what

SmartPRS2 owns the rule and the payroll attendance, SmartEPT owns enforcement, and Caller is the sender. Employees are matched by employee code.

System	Owns (source of truth)	Receives from Caller	Existing feature it plugs into
SmartPRS2	Geofence rule per employee (start point, lat/lng, radius km, Outside = strict / 1 km / 2 km); attendance for payroll	Live sign-in check; login/logout as IN/OUT punches with lat/lng and geo verdict	Field Force → Geofence Rule master, `attendance_logs`, Attendance Report, late-arrival mails, payroll

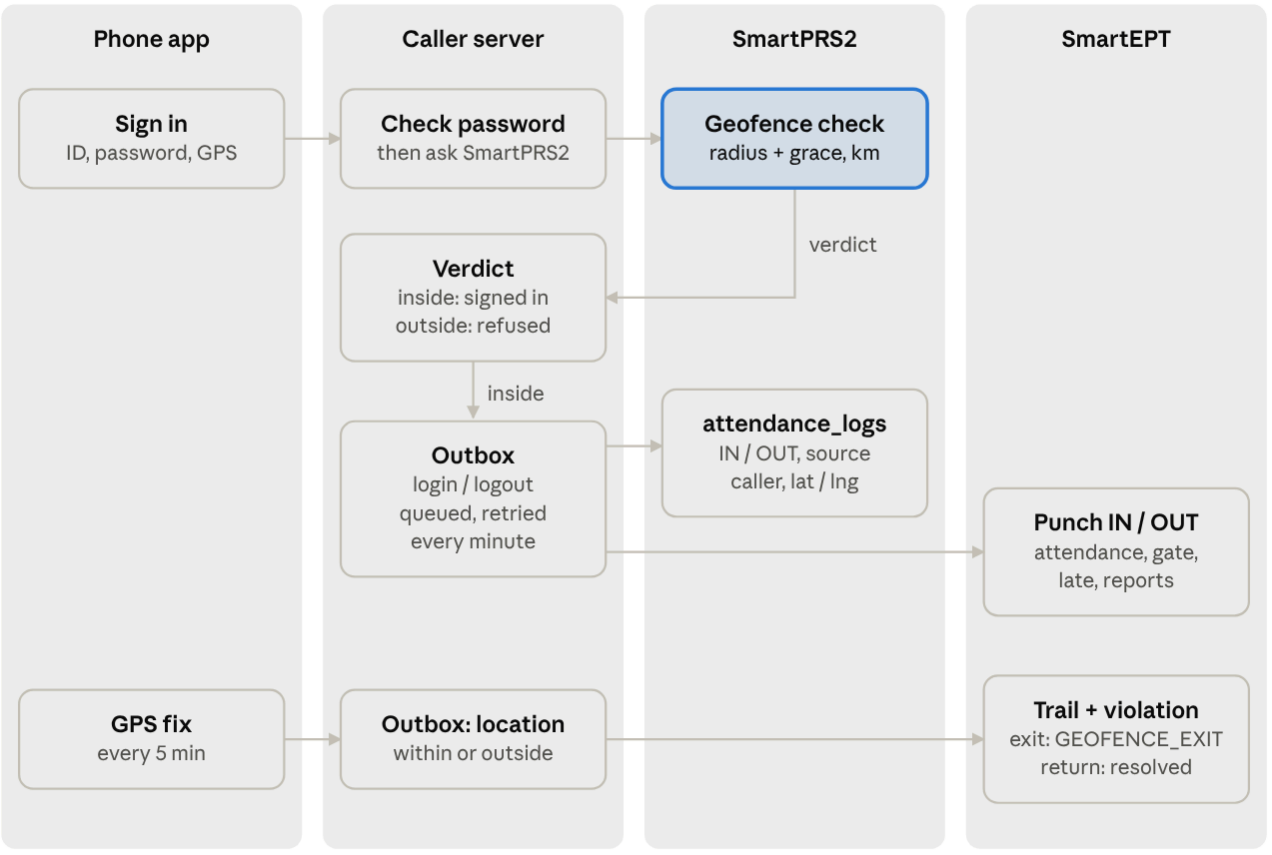
System	Owns (source of truth)	Receives from Caller	Existing feature it plugs into
SmartEPT	Enforcement, monitoring, violations, reports	Login/logout as IN/OUT punches (+ location); every GPS fix as the field trail	Public punch API (gate, attendance merge, derivation), Violations feed (`employee_compliance_events`)
Caller	Phone, sign-in, GPS capture, delivery queue	—	Work sessions, location sync, admin console

The Caller User ID must equal the SmartPRS2 **Emp. Code** and the SmartEPT **Employee Code**. The match is case-insensitive.

Workflow

Sign-in is the only step that waits on another system. Everything after it goes through Caller's outbox, so an offline SmartPRS2 or SmartEPT never blocks the field employee.

SmartPRS2 gates sign-in; the outbox carries every event onward



Delivery runs right after each request; failures retry every minute with back-off (1 to 60 min).

field-force workflow · 4 systems, 3 flows

1. **Sign-in:** the app sends ID, password and a fresh GPS fix. Caller verifies the password, then asks SmartPRS2 whether this point is inside the employee's fence. Outside a blocking fence, sign-in is refused with the distance and limit, and nothing is recorded.
2. **Login and logout:** each one is queued once and delivered twice. SmartPRS2 records it in `attendance_logs` as IN/OUT; SmartEPT records it as a punch and runs its attendance, gate and late rules.
3. **Field trail:** every GPS fix (5 min by default) goes to SmartEPT, marked within or outside the fence. The first fix outside opens a `GEOFENCE_EXIT` violation; the first one back closes it.
4. **Delivery:** the outbox sends right after each request and retries every minute with back-off of 1 to 60 minutes. Each event carries an `external_id`, so a resend never creates a duplicate.

SmartPRS2 changes

SmartPRS2 (`C:\laragon\www\smartprs2` , the live product) gained two API endpoints, built on its existing API keys, geofence master and `attendance_logs` . No existing file's logic was edited.

File	Type	What it does
`app/Services/FieldForceGeofence.php`	New	Evaluates an employee's `geofence_rules` row against a lat/lng: within / outside / no-gps / no-rule, distance and limit in km. Same maths as the in-app self punch (radius + 0 / 1 / 2 km grace).
`app/Http/Controllers/Api/FieldForceApiController.php`	New	`geofenceCheck` (may this employee sign in here?) and `attendance` (writes login/logout as IN/OUT punches).
`routes/api.php`	+2 lines	Registers both routes in the existing `/api/v1` group behind `api-key:ingest` .
`database/migrations/2026_10_06_000100_attendance_logs_geo_columns.php`	New	Makes `attendance_logs.lat` , `lng` , `geo_status` permanent. Before, they were added lazily on the first self punch. Idempotent.
`tests/Feature/CallerFieldForceTest.php`	New	4 tests: fence in/out/no-GPS, no rule, unknown employee, timezone conversion, idempotency, auth.

How the attendance rows behave:

- **Row content:** each event becomes one `attendance_logs` row with `source = caller` , `device_sn = CALLER` , `direction` in/out, `lat` , `lng` and `geo_status` . The Attendance Report, punch pairing, late-arrival mail (`NotifyLateArrivals`) and payroll read it like a biometric punch.

- **Tenant:** comes from the API key, never from the request body, the same rule SBB follows.
- **Geofence verdict:** recomputed by SmartPRS2. The sender's own verdict is not trusted.
- **Time:** must be ISO-8601 with an offset. It is converted to app-timezone wall clock; a time without an offset is rejected rather than guessed.
- **Retries:** a retried event is a no-op through the existing `(tenant_id, external_id)` unique index.

SmartEPT changes

SmartEPT (`C:\laragon\www\smartept`) reuses its public punch API for sign-in and sign-out, and gained a location-trail endpoint whose geofence exits appear as ordinary violations.

File	Type	What it does
`app/Http/Controllers/Api/PublicApiController.php`	Edited (additive)	`POST /api/v1/attendance/punches` also accepts optional `lat` , `lng` , `accuracy_m` , `geo_status` per punch, stored in `biometric_logs.metadata` . Existing callers (SBB) are unaffected.
`app/Http/Controllers/Api/FieldForceController.php`	New	`ingest` stores GPS fixes and raises or resolves violations; `trail` returns a day's fixes.
`routes/api.php`	+2 lines	`POST` and `GET /api/v1/fieldforce/locations` in the existing public `v1` group (`api-key:ingest` / `api-key:read`).
`database/migrations/2026_10_06_000100_create_field_force_locations_table.php`	New	Table `field_force_locations` : company, employee, code, external_id (unique per company), captured_at, lat, lng, accuracy, battery, geo_status, distance_km.
`tests/Feature/CallerFieldForceTest.php`	New	2 tests: location kept on a sign-in punch; exit raises exactly one violation, return resolves it, re-delivery ignored, trail readable.

What existing SmartEPT features now do with Caller data:

- **Attendance, gate, derivation:** a Caller login/logout runs through the same `mergeIntoAttendance` , `processGatePunches` and `deriveAttendance` steps as a biometric punch. Shift-aware check-in/out, late marking and Gate-to-PC treat a field

sign-in as a punch.

- **Violations:** the first fix outside the fence creates an `employee_compliance_events` row: type `GEOFENCE_EXIT`, category `DEVICE`, severity `MEDIUM`, with distance and limit. It shows in the existing Violations feed, reports and alerts. The next fix back inside sets `resolved_at`.
- **Outbound push to SmartPRS2:** unchanged. See Known limits for the double-count case.

Caller changes (supporting)

Caller holds the connection settings and a delivery queue, and checks the fence at sign-in. Without a configured connection it behaves exactly as before.

Area	Change
Database	`integration_connections` (per company: system, base URL, API key encrypted, enabled, block-outside-fence, last status) and `integration_outbox` (pending / sent / failed events with retry time)
`FieldForceIntegration` service	Geofence gate at sign-in, outbox writes on login/logout/location, delivery with back-off, connection test
Sign-in (`SessionService::login`)	Calls the SmartPRS2 check after the password is verified. Outside the fence it refuses with the PRS message; if PRS is unreachable it lets the user in (`unverified`).
Sign-out, location sync	Queue a logout punch; queue each GPS fix for SmartEPT with within/outside against the fence cached at sign-in
Scheduler	`integrations:flush` every minute; also flushes right after each request
Admin console	Administration → **Integrations (PRS / EPT)**: URL, API key, enable, block/record-only, Save & test, Send waiting data now
Android app	Sends a fresh GPS fix (8 s timeout) with the sign-in request

API reference

All calls send `X-Api-Key: <key>` and JSON. The tenant or company always comes from the key.

System	Method & path	Key scope	Purpose
SmartPRS2	`POST /api/v1/fieldforce/geofence/check`	ingest	May this employee sign in here?
SmartPRS2	`POST /api/v1/fieldforce/attendance`	ingest	Login/logout → IN/OUT attendance
SmartEPT	`POST /api/v1/attendance/punches` (existing)	ingest	Login/logout → IN/OUT punch, now with location
SmartEPT	`POST /api/v1/fieldforce/locations`	ingest	GPS trail; raises/resolves `GEOFENCE_EXIT`
SmartEPT	`GET /api/v1/fieldforce/locations?employee_code=&date=`	read	A day's trail

Geofence check (SmartPRS2)

```
// request
{"emp_code": "EMP001", "lat": 17.3850, "lng": 78.4867}
// response
{"ok": true, "employee_found": true, "allowed": false, "status": "outside",
 "distance_km": 3.1, "limit_km": 0.5,
 "fence": {"start": "office", "lat": 17.46, "lng": 78.36, "radius_km": 0.5,
"outside": "strict", "limit_km": 0.5},
 "message": "Outside your allowed area: you are 3.1 km from your office point
(limit 0.5 km)."}
```

`status` is one of `within`, `outside`, `no-gps` (blocked only for strict fences), `no-rule` or `unknown-employee`. The last two are always allowed.

Attendance (SmartPRS2)

```
// request
{"events": [{"external_id": "caller-login-12-20261006093000", "emp_code":
"EMP001", "type": "login",
             "at": "2026-10-06T09:30:00+05:30", "lat": 17.4601, "lng":
78.3601, "accuracy_m": 12}]}
// response — one verdict per event: accepted | duplicate | unknown_employee
| rejected
{"ok": true, "results": [{"external_id": "caller-login-12-20261006093000",
"status": "accepted", "geo_status": "within"}]}
```

Punch with location (SmartEPT, existing endpoint)

```
{"punches": [{"employee_code": "EMP001", "punch_type": "IN", "punched_at":
"2026-10-06 09:30:00",
              "external_id": "caller-login-12-20261006093000", "source":
"CALLER",
              "lat": 17.4601, "lng": 78.3601, "geo_status": "within"}]}
```

Location trail (SmartEPT)

```
// request
{"locations": [{"external_id": "caller-loc-881", "employee_code": "EMP001",
"captured_at": "2026-10-06T11:05:00+05:30",
                "lat": 17.46, "lng": 78.388, "accuracy_m": 12, "battery_pct":
64,
                "geo_status": "outside", "distance_km": 2.97, "limit_km":
0.5}]}
// response (202)
{"ok": true, "received": 1, "stored": 1, "unknown_employee_codes": []}
```

Setup and go-live checklist

Setup takes about 15 minutes per company. Do it on a test copy before touching a live client.

- ☐ **SmartPRS2:** run `php artisan migrate` (adds the geo columns if missing).
- ☐ **SmartPRS2:** Time & Attendance → API Keys → create a key with scope **ingest** for the right workspace.
- ☐ **SmartPRS2:** Field Force → Geofence Rule → set start point, lat/lng, radius and Outside mode for each field employee.
- ☐ **SmartEPT:** run `php artisan migrate` (creates `field_force_locations`).
- ☐ **SmartEPT:** Integrations → API keys → create a key with scopes **ingest** and **read**.

- [ ] **Caller:** in `backend`, run `php artisan migrate`. Keep the scheduler running (`php artisan schedule:work` or the Windows task).
- [ ] **Caller:** in `admin`, run `npm install` (first time only), then `npm run build`.
- [ ] **Caller admin:** Administration → Integrations (PRS / EPT) → enter both URLs and keys → **Save & test**. Each card should say "Connected to <company>".
- [ ] **Employee codes:** check that each Caller User ID matches the PRS Emp. Code and the EPT Employee Code.
- [ ] **Double counting:** decide whether SmartEPT should keep pushing attendance to SmartPRS2 for field staff (see Known limits).
- [ ] **Android:** rebuild and install the app (Run ▶). Sign in inside and outside the fence and check both outcomes.

Testing and verification

The new tests pass in all three systems, and the existing tests in the same suites still pass, run against clean copies with an in-memory SQLite database.

System	Test file	Result	What it proves
SmartPRS2	`tests/Feature/CallerFieldForceTest.php`	4 tests, 30 assertions, pass	Inside/outside/no-GPS verdicts; no-rule and unknown employees allowed; login/logout stored as in/out in IST; retry is a duplicate; no key = 401
SmartEPT	`tests/Feature/CallerFieldForceTest.php`	2 tests, 12 assertions, pass	Location kept on the punch; one violation per exit, resolved on return; re-delivery ignored; trail readable
SmartEPT	`tests/Feature/SbbPunchIngestTest.php` (existing)	18 tests, pass	The edited punch endpoint still behaves for the biometric bridge
Caller	`tests/Feature/FieldForceIntegrationTest.php`	3 tests, 30 assertions, pass	Outside = blocked with nothing recorded; inside = PRS attendance, EPT punch and outside location delivered; PRS outage = sign-in allowed, retry scheduled

System	Test file	Result	What it proves
Caller admin	TypeScript + Vite build	Pass	New Integrations page compiles

One bug surfaced on your PC after the tests: the Integrations page failed on MySQL 8 because `system` is a reserved word there. It is fixed. The tests use SQLite, which does not reserve that word, so they did not catch it.

Not yet verified on a real phone: the Android sign-in with GPS. It needs a rebuild and a field test.

Known limits and next steps

The main open decision is how to avoid counting field staff twice in SmartPRS2. Everything else is a later improvement.

#	Limit	Effect	Suggested fix
1	SmartEPT may also push attendance to SmartPRS2 (Integration target on `attendance.daily` / `attendance.punch`)	SmartPRS2 receives the same login twice, once as source `caller` and once as `smartept`	For field staff, keep only the direct Caller → SmartPRS2 path, or turn the Caller → SmartPRS2 attendance off
2	The fence used for GPS fixes is cached at sign-in for 14 h	A rule changed mid-day applies from the next sign-in	Acceptable; re-check on each heartbeat if needed
3	No SmartEPT screen draws the trail on a map	Trail visible in Caller → Status & location and the EPT read API; exits appear as violations	Add a map page in the SmartEPT admin UI
4	SmartPRS2 geofence maths lives in two places (`FieldForceGeofence` and `AttendanceReportController::checkGeofence`)	A rule change must be made twice	Point the controller at the new service the next time it is edited
5	Sign-in fails open when SmartPRS2 is unreachable	During an outage, staff outside the fence can sign in; their attendance is marked `unverified`	Keep, or make it a per-company setting if a client needs strict mode

#	Limit	Effect	Suggested fix
6	Integration settings use existing permissions (view: System health; edit: Configure calling)	No separate role for integrations	Add an `integrations.manage` permission if needed

Next: run the go-live checklist on one test company, then a field test with the OPPO phone.